

The main objectives of this analysis are:

1. To obtain the aerodynamic stability derivatives of the aircraft.
2. To determine the longitudinal and lateral-directional stability characteristics.
3. To determine the linearized state-space representation of the aircraft.
4. To calculate the eigenvalues of the linearized dynamic system.
5. To identify the aircraft's principal dynamic modes.
6. To evaluate the damping, natural frequency, and stability of each mode.

No control-surface actuation or feedback control system was introduced in this analysis. Therefore, the calculated eigenvalues represent the uncontrolled aircraft dynamics at the selected operating condition.

1 Aircraft Configuration and Reference Data

The AVL model represents the complete aerodynamic configuration of the aircraft, consisting of the main wing, horizontal tail, and vertical tail.

The aerodynamic reference quantities used in the AVL model are summarized in Table 1.

Table 1: Aircraft reference quantities used in AVL.

Parameter	Value
Reference area, S_{ref}	0.08001 m ²
Reference chord, c_{ref}	0.12969 m
Reference span, b_{ref}	0.63000 m
X_{ref}	0.07879 m
Y_{ref}	0.00000 m
Z_{ref}	0.00697 m
Air density, ρ	1.225 kg/m ³
Gravity, g	9.81 m/s ²
Aircraft mass, m	0.514 kg

The total mass was obtained from the complete mass breakdown:

$$\begin{aligned}
 m &= 0.125 + 0.018 + 0.010 + 0.134 \\
 &\quad + 0.075 + 0.028 + 0.124 \\
 &= 0.514 \text{ kg.}
 \end{aligned} \tag{1}$$

The calculated center of gravity is approximately

$$X_{CG} = 0.07880 \text{ m,} \tag{2}$$

$$Y_{CG} \approx 0, \tag{3}$$

and

$$Z_{CG} = 0.006975 \text{ m.} \quad (4)$$

2 Mass and Inertia Properties

The mass and inertia properties supplied to AVL were obtained from the aircraft mass model.

The principal inertia values used in the dynamic analysis were

$$I_{xx} = 0.003232 \text{ kg m}^2, \quad (5)$$

$$I_{yy} = 0.004960 \text{ kg m}^2, \quad (6)$$

$$I_{zz} = 0.008001 \text{ kg m}^2. \quad (7)$$

The corresponding inertia tensor was

$$\mathbf{I} = \begin{bmatrix} 0.003232 & 0 & 1.89 \times 10^{-5} \\ 0 & 0.004960 & 0 \\ 1.89 \times 10^{-5} & 0 & 0.008001 \end{bmatrix} \text{ kg m}^2. \quad (8)$$

These values were used by AVL in constructing the linearized aircraft equations of motion.

3 AVL Analysis Setup

The aircraft was analyzed using AVL version 3.37.

The aerodynamic model consisted of:

- Main wing,
- Horizontal tail,
- Vertical tail.

The model contained 83 aerodynamic strips and 809 vortex elements.

The following operating condition was selected for the dynamic analysis:

$$V = 20.0 \text{ m/s}, \quad (9)$$

with

$$\rho = 1.225 \text{ kg/m}^3. \quad (10)$$

The corresponding Run Case 1 solution produced a required lift coefficient of

$$C_L = 0.25723. \quad (11)$$

4 Aerodynamic Operating Point

At the selected operating condition, AVL returned the following aerodynamic solution:

Table 2: Aerodynamic operating point obtained from AVL.

Quantity	Value
Velocity, V	20.0 m/s
Angle of attack, α	-8.303°
Lift coefficient, C_L	0.25723
Drag coefficient, C_D	0.01696
Pitching moment coefficient, C_m	+0.21928
C_X	-0.05393
C_Z	-0.25208
Induced drag coefficient, C_{D_i}	0.016958
Span efficiency, e	0.2679

4.1 Operating Point Limitation

An important distinction must be made between the lift condition imposed during the AVL analysis and a complete longitudinal trim condition.

The calculated pitching moment coefficient was

$$C_m = +0.21928. \quad (12)$$

Since

$$C_m \neq 0, \quad (13)$$

the selected operating point is not a complete longitudinal pitching moment equilibrium.

Therefore, the eigenvalues presented in this report should be interpreted as the linearized dynamic characteristics of the aircraft around the selected AVL operating point.

This limitation is explicitly stated because the purpose of the present analysis is to extract and evaluate the aircraft's dynamic modes, while the final trimmed aerodynamic condition is investigated separately.

5 Static Stability Derivatives

AVL was used to calculate the aircraft stability derivatives.

The resulting derivatives are presented in Table 3.

Table 3: Static stability derivatives obtained from AVL.

Derivative	Value
C_{L_α}	4.757761
C_{L_β}	0
C_{Y_β}	-0.246263
C_{l_β}	-0.081797
C_{m_α}	-0.895599
C_{m_β}	0
C_{n_β}	0.112811

The negative pitching moment derivative

$$C_{m_\alpha} = -0.895599 \quad (14)$$

indicates positive longitudinal static stability.

The lateral-directional derivatives also indicate favorable static stability characteristics:

$$C_{l_\beta} = -0.081797, \quad (15)$$

and

$$C_{n_\beta} = +0.112811. \quad (16)$$

These results indicate restoring rolling and yawing moments following small sideslip disturbances.

6 Dynamic Stability Derivatives

The dynamic stability derivatives calculated by AVL are shown in Table 4.

Table 4: Dynamic stability derivatives obtained from AVL.

Derivative	Value
C_{L_p}	0
C_{L_q}	9.007163
C_{L_r}	0
C_{Y_p}	-0.236351
C_{Y_q}	-0.000002
C_{Y_r}	0.309552
C_{l_p}	-0.426048
C_{l_q}	0
C_{l_r}	0.136908
C_{m_p}	0
C_{m_q}	-16.254972
C_{m_r}	0
C_{n_p}	0.016380
C_{n_q}	-0.000002
C_{n_r}	-0.155319

The roll damping derivative

$$C_{l_p} = -0.426048 \quad (17)$$

indicates a restoring aerodynamic moment opposing roll rate.

Similarly, the pitch damping derivative

$$C_{m_q} = -16.254972 \quad (18)$$

indicates strong aerodynamic damping in pitch.

The yaw damping derivative is

$$C_{n_r} = -0.155319, \quad (19)$$

which also contributes to damping of the lateral-directional response.

7 Neutral Point and Static Margin

The neutral point calculated by AVL was

$$X_{NP} = 0.103210 \text{ m}. \quad (20)$$

The aircraft center of gravity was

$$X_{CG} = 0.07880 \text{ m}. \quad (21)$$

The static margin is therefore

$$SM = \frac{X_{NP} - X_{CG}}{c_{\text{ref}}}. \quad (22)$$

Substituting the measured values gives

$$SM = \frac{0.103210 - 0.07880}{0.12969} \approx 0.188. \quad (23)$$

Thus,

$$\boxed{SM \approx 18.8\%} \quad (24)$$

which indicates a positive static margin.

This result agrees with the negative value of C_{m_α} and confirms positive longitudinal static stability.

8 Linearized State-Space Model

AVL generated the linearized equations of motion in state-space form:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}. \quad (25)$$

The state vector used by AVL was

$$\mathbf{x} = \begin{bmatrix} u & w & q & \theta & v & p & r & \phi & x & y \\ z & \psi & & & & & & & & \end{bmatrix}^T. \quad (26)$$

The system matrix obtained directly from AVL was

$$\mathbf{A} = \begin{bmatrix} -0.3538 & -0.9998 & 2.7917 & -9.8100 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & & & & & & & & \\ -2.2079 & -8.6195 & 18.2728 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & & & & & & & & \\ 7.1011 & -22.1349 & -24.5711 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & & & & & & & & \\ 0 & 0 & 1.0000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & & & & & & & & \\ 0 & 0 & 0 & 0 & -0.4689 & -3.0633 & -19.5583 & 9.8100 & 0 & 0 \\ 0 & 0 & 0 & 0 & -10.9110 & -22.1584 & 9.3661 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 9.1177 & 0.6975 & -4.1497 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0000 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1.0000 & 0 & 0 & -2.8883 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.0000 & 0 & 0 & 2.8883 & 0 & 0 \\ 0 & 19.7903 & & & & & & & & \\ 0 & 1.0000 & 0 & -19.7903 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 1.0000 & 0 & 0 & 0 \\ 0 & 0 & & & & & & & & \end{bmatrix}. \quad (27)$$

The eigenvalues of this matrix provide the natural dynamic modes of the aircraft.

9 Eigenvalue Analysis

For a linearized dynamic system, each eigenvalue may be written as

$$\lambda = \sigma \pm i\omega, \quad (28)$$

where σ is the real part and ω is the imaginary part.

The sign of σ determines stability:

$$\sigma < 0 \quad \Rightarrow \quad \text{stable mode}, \quad (29)$$

while

$$\sigma > 0 \quad \Rightarrow \quad \text{unstable mode}. \quad (30)$$

The eigenvalues obtained directly from AVL are shown in Table 5.

Table 5: Eigenvalues obtained from AVL Run Case 1.

Mode	Eigenvalue	Type
1	-23.800234	Real
2-3	$-1.5364187 \pm 13.028574i$	Complex pair
4-5	$-16.775574 \pm 17.971889i$	Complex pair
6	+0.096084811	Real
7-8	$+0.0033943544 \pm 1.3366623i$	Complex pair

10 Dynamic Mode Identification

The eigenvalue spectrum corresponds to five principal aircraft dynamic modes:

1. Roll subsidence,
2. Dutch roll,
3. Short period,
4. Spiral mode,
5. Phugoid.

Complex conjugate eigenvalues correspond to oscillatory modes, whereas real eigenvalues correspond to non-oscillatory modes.

10.1 Roll Subsidence

The first eigenvalue is

$$\lambda_{\text{roll}} = -23.800234. \quad (31)$$

The negative real eigenvalue indicates a stable, non-oscillatory roll subsidence mode.

The characteristic time constant is

$$\tau_{\text{roll}} = \frac{1}{|\sigma|} = \frac{1}{23.800234} \approx 0.0420 \text{ s}. \quad (32)$$

The corresponding half-life is

$$t_{1/2} = \frac{\ln 2}{23.800234} \approx 0.0291 \text{ s.} \quad (33)$$

Therefore, the roll disturbance decays very rapidly.

10.2 Dutch Roll

The Dutch-roll eigenvalue pair is

$$\lambda_{\text{DR}} = -1.5364187 \pm 13.028574i. \quad (34)$$

The natural frequency is calculated from

$$\omega_n = \sqrt{\sigma^2 + \omega^2}. \quad (35)$$

Therefore,

$$\omega_n = \sqrt{(-1.5364187)^2 + (13.028574)^2} \approx 13.119 \text{ rad/s.} \quad (36)$$

The damping ratio is

$$\zeta = -\frac{\sigma}{\omega_n} \approx 0.117. \quad (37)$$

The oscillation period is

$$T = \frac{2\pi}{|\omega|} = \frac{2\pi}{13.028574} \approx 0.482 \text{ s.} \quad (38)$$

The Dutch-roll mode is therefore stable but lightly damped.

10.3 Short-Period Mode

The short-period eigenvalue pair is

$$\lambda_{\text{SP}} = -16.775574 \pm 17.971889i. \quad (39)$$

The natural frequency is

$$\omega_n = \sqrt{(-16.775574)^2 + (17.971889)^2} \approx 24.583 \text{ rad/s.} \quad (40)$$

The damping ratio is

$$\zeta = -\frac{-16.775574}{24.583} \approx 0.682. \quad (41)$$

The oscillation period is

$$T = \frac{2\pi}{17.971889} \approx 0.350 \text{ s.} \quad (42)$$

The short-period mode is therefore stable and strongly damped.

10.4 Spiral Mode

The sixth eigenvalue is

$$\lambda_{\text{spiral}} = +0.096084811. \quad (43)$$

Because the real part is positive, the spiral mode is unstable.

The disturbance amplitude behaves approximately as

$$x(t) \propto e^{0.096084811t}. \quad (44)$$

The corresponding doubling time is

$$t_2 = \frac{\ln 2}{0.096084811} \approx 7.22 \text{ s}. \quad (45)$$

Thus, the amplitude associated with the spiral mode doubles approximately every 7.22 seconds in the absence of corrective control input.

This is the most significant dynamic instability identified in the present analysis.

10.5 Phugoid

The phugoid eigenvalue pair is

$$\lambda_{\text{PH}} = +0.0033943544 \pm 1.3366623i. \quad (46)$$

The natural frequency is

$$\omega_n = \sqrt{(0.0033943544)^2 + (1.3366623)^2} \approx 1.33666 \text{ rad/s}. \quad (47)$$

The damping ratio is

$$\zeta = -\frac{0.0033943544}{1.33666} \approx -0.00254. \quad (48)$$

The negative damping ratio indicates that the mode is very slightly unstable.

The oscillation period is

$$T = \frac{2\pi}{1.3366623} \approx 4.70 \text{ s}. \quad (49)$$

The amplitude doubling time is

$$t_2 = \frac{\ln 2}{0.0033943544} \approx 204.2 \text{ s}. \quad (50)$$

Therefore, although the phugoid is mathematically unstable, its growth rate is extremely small compared with that of the spiral mode.

11 Dynamic Mode Summary

The principal dynamic characteristics are summarized in Table 6.

Table 6: Summary of the aircraft dynamic modes.

Mode	σ	ω	ω_n	ζ	Stability
Roll subsidence	-23.800	0	23.800	1.000	Stable
Dutch roll	-1.536	± 13.029	13.119	0.117	Stable
Short period	-16.776	± 17.972	24.583	0.682	Stable
Spiral	+0.0961	0	0.0961	–	Unstable
Phugoid	+0.00339	± 1.337	1.337	-0.00254	Very slightly unstable

12 Overall Stability Assessment

The eigenvalue analysis reveals three stable dynamic modes and two unstable modes.

The roll subsidence mode is highly stable, with a very short time constant. The short-period mode is also strongly damped, with a damping ratio of approximately

$$\zeta_{\text{SP}} \approx 0.682. \quad (51)$$

The Dutch-roll mode is stable but lightly damped:

$$\zeta_{\text{DR}} \approx 0.117. \quad (52)$$

The principal instability is the spiral mode, which has a positive real eigenvalue of

$$\sigma_{\text{spiral}} = 0.0961 \text{ s}^{-1}. \quad (53)$$

Its doubling time of approximately 7.22 seconds indicates a slowly diverging lateral-directional response.

The phugoid is also mathematically unstable, but its growth rate is much smaller:

$$\sigma_{\text{phugoid}} = 0.00339 \text{ s}^{-1}. \quad (54)$$

Its corresponding doubling time is approximately 204 seconds. Consequently, the phugoid instability is extremely weak.

13 Comparison with Static Stability

The dynamic results are generally consistent with the static stability characteristics.

The negative pitching-moment derivative

$$C_{m_\alpha} = -0.895599 \quad (55)$$

and the positive static margin

$$SM \approx 18.8\% \quad (56)$$

indicate positive longitudinal static stability.

This agrees with the stable short-period mode.

Similarly,

$$C_{l_\beta} = -0.081797 \quad (57)$$

and

$$C_{n_\beta} = +0.112811 \quad (58)$$

indicate favorable static lateral-directional stability.

However, the unstable spiral eigenvalue demonstrates that positive static stability does not necessarily guarantee complete dynamic stability.

This distinction is important in aircraft design: static stability describes the initial restoring tendency following a disturbance, whereas dynamic stability describes the subsequent time evolution of that disturbance.

14 Conclusions

The AVL dynamic stability analysis provided the linearized dynamic characteristics of the aircraft at the selected operating condition.

The calculated eigenvalues were

$$\begin{aligned} \lambda_1 &= -23.8002, \\ \lambda_{2,3} &= -1.5364 \pm 13.0286i, \\ \lambda_{4,5} &= -16.7756 \pm 17.9719i, \\ \lambda_6 &= +0.09608, \\ \lambda_{7,8} &= +0.003394 \pm 1.33666i. \end{aligned} \quad (59)$$

The aircraft therefore exhibits:

- stable roll subsidence,
- stable but lightly damped Dutch roll,
- stable and strongly damped short-period motion,
- an unstable spiral mode,
- a very weakly unstable phugoid mode.

The spiral mode represents the primary dynamic stability concern because it has the largest positive real eigenvalue and therefore the fastest growth among the unstable modes.

The phugoid instability is considerably weaker and develops over a much longer time scale.

The static stability results are favorable, with a positive static margin of approximately 18.8% and a negative C_{m_α} .

Finally, the interpretation of the eigenvalues must account for the fact that the selected AVL operating point satisfies the required lift condition but does not represent a complete longitudinal trim condition, as indicated by

$$C_m = +0.21928. \quad (60)$$

Therefore, the results presented here should be regarded as the linearized dynamic characteristics of the aircraft at the analyzed AVL operating point. A final dynamic stability assessment at a fully trimmed flight condition can be performed subsequently if required.